Inês Hipólito ~ Active Inference Insights 004 ~ Markov Blankets, Modularity, Enactivism

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welcome everybody to the fourth episode of active inference insights I'm your host Daris Pary Wayne and today I have the pleasure of Hosting inz hippolito inz is a lecture of philosophy of artificial intelligence and mqu University in Australia our research focuses on 4E cognition which holds that cognition is embodied embedded enacted and extended beyond the brain and she studies how this framework can help us understand and build other systems that exhibit a form of augmented Mission including brain computer interfaces neurot Technologies and smart environments in addition to this she co-founded the international Society of the philosophy of the Sciences of the mind with gutier picini and she is currently the ethics uh ethics advisor um in artificial intelligence versus the cognitive Computing company which implements active inference into artificial intelligence agents um inz thank you so much for joining me it's an absolute pleasure thank you so much for having me uh the pleasure is mine and it's um we were just discussing off a that you know we're doing this across many oceans so Nez is in Australia I'm in London um but we managed to sync it up which is which is fantastic exent so I wanted to start with markof blankets um all things start with Mar Markov blankets um Mark of blankets really play a fundamental role in active and in the framework that we apply to artificial intelligence to cognition more generally and also just the underlying physics and math of back to inference and a lot of your work has focused on Mark of blankets and the scaling up of Mark of blankets so maybe a good place to start for our audience would be if you could just give a simple overview of exactly what a mark of blanket is and then we can go from that yeah fantastic uh yeah you're absolutely right when you say that everything starts with the mark of blanket everything that is a thing um is has got a mark of blanket and that does not necessarily mean um that a mark of blanket is going to correspond to a certain kind of like physical boundary that one can see uh right a mark of blanket um is a theoretical formal construct that what it does is it tells you um how or what things are going to be influencing a certain state to be the case right so it's it is a statistical Tool uh precisely to understand something really important uh which is how a certain thing is coupled with within or with its environment um so everything does start everything has its markco blanket does start with the markco blanket because it is a way in which we can capture two very important aspects um and these two aspects um are that um a thing um typically an open system as the thing um is um both thermodynamically open and operationally closed so this is quite crucial um because it is what allows the thing um to persist not to perish not to succumb to the second law of Thermodynamics so Mark of blanket is really become uh very useful because to my understanding um they allow us to capture simultaneously these two features being both operationally open oh sorry operationally closed and thermodynamically open because what happens is the system is situated in the environment in a way that it is open to the environment it must be coupled with the environment otherwise it is uh perishing um it will not maintain itself but it also has a certain boundary that allows it to maintain its coherent for so the markco blankets become really interesting because they allow us to capture those two dimensions of um an open system and what is an open system an open system is mostly everything that we see uh we don't really come across uh closed systems excellent yeah that's great um I'm wondering whether you could give us a sort of overview of exactly how that coupling works so people may have seen that the mark of blanket is essentially composed of active States and sensory States um that makes sense if yet if we're talking kind of about physical boundaries like actuators arms legs and so on but on the understanding The Mark of blanket is a statistical entity what does um what what what really are active States and sensory States um in the theoretical way well there are many ways to answer this question um it will depend on how much you want to connect um then the active inference framework and formalism uh with other uh theories that we find in physics or in Neuroscience uh so there are many different ways in which one can answer um this question um but for the sake of remaining um with the kind of like more scale-free application of markco blankets and not reducing to immediately uh to the brain um I will uh keep the formalism as something that we can apply to understand the statistical relations or influences um between a system an open system and its environment um right so then what you have is you have the possibility of understanding how a certain system is in a particular state so you understand that and then and then what you can also understand is you can make predictions about future States in which you may find the system by virtue of using this kind of for formulas right uh so then what you have is you employ or Define that the level or aspects um of scientific interest are going to be your internal States right and then your next question is going to be okay how do I find this system in this PR in this precise uh time and space in these particular States what are the the the the states uh in which the system is situated that are going to be crucial for me to find the system in this particular State and therefore I understand um the system and potentially ideally I can make some uh nice predictions about future States so this will be our internal States so then what we need to we need to get on is then we need to get to Define what are the external States and those are like I said the states that are going to be influential uh for immediately influential for um the system to be in a specific kind of like internal States now this is all done the next step is to understand that internal and external States they mutually influence one another that's what it means about something being an open system right is that there is exchange of mattera and energy such that the system is remaining in a certain State and for example is not um going into into a face space um uh such as like perishing or or towards um towards uh death um so then um what what you get is an understanding of um what I was just saying which is how the system is coupled and in this permanent um continuous exchange of matter and energy in order to remain alive and push back on the second LA themodynamics right so you get this operationally closed and open system at the same time that is uh very well captured by these internal and external States and I did say that uh they influence one another but it is important um to understand that they do not directly influence one another they need a second set of States um so basically internal and external states are conditionally in opin it and they will require the intervention of a second um set of states which is very much what you asked uh the active and sensory States right so these are going to be the ones that are doing the direct communication if you will between each other where um hopefully and ideally we have um a balanced influence between sensory and active States right and these um direct influences between um sensory and active states are going to play a crucial role in order for us to mathematically trap down and understand the coupling of the system in the environment which is the system being internal States the environment being external States so this is just like the overview of uh the mark of blanket and the active inference formalism that we could apply to anything that is an open system wonderful yeah that's great that's a really useful explication of the framework um I think it's as I kind of hinted to in my initial question it's it's rather easy to think of it as a physical boundary so we can think of cell boundaries or the blood Bane barrier or the boundaries of my body and then when we do that the notion of sense reactive States really becomes intuitive in the sense that in the case of my body all it is is an action perception Loop and that's being known to be critical to intelligence or cognition for a very long time so that said there's also a very strong position which is that it's actually just a statistical description of the way things act the way things persist through time that leads me to ask the question of should we consider the markco blanket to be in uh interpreted in an instrumentalist way in the sense that it's kind of a way it's a tool that we can use in modeling objects open systems or is it or should we be inter in it in a realistic way I.E there is actually something uh that is ontologically a mark of blanket that is truly out there as a markco blanket um embedded within lots of other markco blankets of course yes I'm really glad that you ask that uh question it's a Live question um and um there are many ways to answer um this question so the way that I that I that I that I decided to answer the previous question was was in a way that one understands that um a mark of blanket is a tool is a statistical tool it's a formalism it's a mathematical construct right and I also did not go immediately into the brain uh or an organism with actions and Sensations because I'm trying to avoid uh precisely um the literature that tends to reduce the framework to thinking that active States and sensory states are going to be action and sensation right that is not the case right um so a mark of blanket formalism from active inference can be applyed to anything that is an open system and there are many um examples of this in the literature such as for example I can think of um the example of when you have a set of um of pendulums in a room or a set of clocks in the same room what's going to happen is uh that eventually they will synchronize right and you can use this framework to capture to predict that synchronization right so it is important to understand that this is a mathematical construct that you can employ to precisely understand patterns of behavior in the natural world right so we've done that um we understood that so then we can think about it you can use this uh framework to develop for example simulation models which allow you to understand the phenomenon of scientific interest as well as hopefully it's a good model is going to give you some prediction power over future states of the thing which doesn't have to be uh anything to do with cognition can be something um uh that is situated in the natural world right then a certain phenomenon happens specifically in cognitive science when we notice that specifically in cognitive science even if you apply a certain computational framework doesn't have to be active inference a certain computational framework is applied to model simulate for ex and get some predictive power about for example black holes let's say black holes right uh and they use that kind of like mathematical framework and then you employ the same kind of mathematical framework the same kind of like ationalism to understand the brain because after all both a black hole and uh a brain are complex systems what does that mean it means that it is extremely difficult for us to model it why is it extremely difficult to model it because it is very high in dimension so it's really hard for us we don't have the computational power to to model it which is why we need to incorporate something called uh dimensionality reduction because we don't have the tractability to to Pro precisely do that now keeping up with what what I was saying in cognitive science specifically a phenomenon happens so imagine this we use the same computational um strategy to model things from the natural world all the way to the brain right and then what happens is that we would never say that the brain had sorry that the the the black hole or something that is not uh part of uh a living system would have uh specifically a mark of blanket or would engage in literally active inference we would understand very clearly that this is computational strategy that we are developing and employing for epistemic purposes to understand the phenomenon and to get some predictive power that's what we're doing and we are happy with it if we get some predictive power that's success in science and then one once we do this move and apply exactly the same formalisms into for example the brain or an organism right we tend to anthropomorphize it right so we tend to say okay so then the brain is acting precisely literally by engaging in active inference or in other uh Realms engaging in predictive coding right so we lose track of understanding that okay these were the tools that we started off with and we were very happy when we were modeling black holes with these tools we were happy that we had some predictive power but now it's not enough we won't more well now we want to go from The epistemic Virtue into an ontological predictor now we want to say that the thing that we are modeling trying to understand developing um uh models for epistemic gain is precisely has the properties or the ontological properties that the model the computational model has right and that's when you get into making a realist or taking a realist stance on and upon active influence right you don't have to you can do it but then you're going to have to defend that position in a way that okay you're going to have to answer so if you are if you are applying the question that you need to answer then is if you're applying exactly the same framework to model and predict black holes that you want to do the same with brains do you get to say that brains engage literally in active inference in ways that black holes don't and why so that's the realism take and the problems that you find and what where you would uh be led to or you remain in the more cautious kind of position which is on the instrumentalist or non-realist position in which case the position is that all things in the world with with with the real existence right and those things behave in ways that we would develop models to understand that behavior in terms of patterns patterns of behavior the things exist in themselves they interact with the world in themselves the patterns of behavior are our scientific constructs so then we talk and then we can start talking about okay so I can pattern behavior through for example employing developing a model on the basis of Maco blankets active inference and then I pattern the behavior what did I did what I did was I found patterns and therefore an explanation for a Behavior Uh and this explanation is something that I didn't have before right right right right well I'm going to take us towards some realist position now because I think people will find it interesting about the the ontological claims ontological implications uh that act of inference has especially within the cognitive science and the Neuroscience community and I guess when you're taking a real stance on the Markov blanket that there are these structures which have some kind of um embedded nature in the brain right and and people have probably heard frisen talk about how this is actually reflected in the phen phenotypic nature of the brain and its hierarchies um it makes me think about modularity so mod it for people who don't know was introduced by Jerry Fodor um in 1983 and the basic idea is that modules are these kind of um innate neural structures which have kind of very specific functions and fdor in 83 laid out sort of certain category certain functions that they a modular system would have to um you know would have to obey by so domain specificity the fact that modules have to operate on a very specific type of input um there's also informational encapsulation this idea that like at least how he portrayed it in 1983 that these modules are encapsulated with respect to high orderer beliefs that that notion got sort of you know weakened over time let's say and then there were further about the actual neural architecture would have to be involved in mod uh modules but anyway people can get the basic idea um of a module we might have a visual module a hearing module a module for language right which would yeah which is idea clearly taken up um or that tronsky found a lot of joy in um all that to say what does active influence have to say about this picture of the Mind as being modular to any interesting degree and does this notion of kind of at least for predictive coding position position a hierarchy in which you have feedback loops in which the top down affects the bot the top layers affect the bottom layers and vice versa does that actually really have anything to say about modularity and more generally the architecture of the brain yes um well I'm really glad that you you're taking us there we do have um a couple of papers that do um engage a little bit with uh the modularity of the Mind one of them is Mark of blankets in the brain and that paper and the the one is parcels and particles um and um in there we do briefly mention um reasons why thinking uh or patterning the the brain Behavior the brain activity in terms of markco blankets would be useful for many reasons uh for for for for applications that go all the way to DCM uh dynamical caal modeling in in Neuro Imaging but I'm going to set that to the side and I'm going to specifically speak to uh the modularity of the mind you very well said um the modularity of the mind is is one of the is one of those um theories that we've been having in philosophy of mind that has been very foundational for cognitive science and for the ways in which we ask questions for the ways in which we look for things in the brain um and has been uh as influential as it has been refuted uh so very few people today I can think of three people people uh that would endorse good old school fodorian modularity of the mind and there are re there are philosophical reasons for that and there are scientific reasons for that philosophically uh there's been made very substantial solid strong arguments against it and uh scientifically um evidence shows neuroplasticity shows neural reuse in ways that would contradict uh the modularity of the mind I will make here just a recting of a little bit of history of philosophy of mind just um before I get into U the active inference um understanding of um cognitive SLB brain um activity and Dynamics just to say that it's interesting to see that we have this tendency so I'll do a little bit of philosophy of Mind here we have been having this tendency in philosophy of mind or at least analytic philosophy of mind um which is to look at and this is a point that has been made by many um influential philosophers uh for example Francis Egan um she has made this very uh clever point and I think she's absolutely right when uh we look into World philosophers of Mind look into what is happening in computer science and then we adopt the best computer science at the time to establish a kind of like a metaphor to how the mind supposedly work at the time as you said in 83 uh Jerry F looked at um uh touring machines and in he and he asked the question what would happen if one would conceive the mind as if it were a touring machine and then he writes the book right what happened is that what starts off as being like sort of like metaphorical thinking right it eventually starts becoming a very ontological uh claim that the mind is ontologically modular and that is very um it's very important for us to look back in history so that we know what to do and what not to do in the future right because what happens is then you set off cognitive science to look for those modules in the brain as you set up experiments as you set up uh uh uh simulations computational simulations you set off to look for and find those very um then you have all of those sets of properties that are supposed to be the model so there's the automaticity there's the domain specificity there's there's the information encapsulation there's the top down information oh sorry uh the B information going all the way up to cognition cognition mean so all of that framework that was a theoretical framework adopted from computer science then is defining the whole cognitive science field and what we should look for right so it's quite interesting and important for us to know our history to know how things really happen in a way that well you look back and and you see how how it then science came back and said well it kind of actually doesn't make sense that the mind is modular in that way uh so not only philosophically we have refuted it and very u in ways that we have forced the modularity of the mind to reinvent itself to revise itself to survive really and what happened was um because the problems were coming to arise from philosophy and from from from Neuroscience what happened was we started having to revise the modularity of the mind because the noveles were coming up but we did not want to let go of the modularity of the mind so we start revising the basic pillars and then we start having to let go of basic pillars and that's when you have massive modularity by petas for example and still um challenges continue to come up right so then you start going into this other uh this other field which was known at the time um cognitive penetrability so now what you allow is you you you you revise modularity to the point that it's not just bottom up information going up but now you will allow for bottom up and top down that's cognitive penetrability right and then it comes up um around 2010 and stays until 2020 still is a little bit here because as we um sort of like um explored a little bit in a in a in a special issue that we we we just had out um this year with Michael kof it's a special issue in Consciousness and cognition and it's about uh the bean brain and how whether how or whether uh the visian brain predictive coding is compatible to the modularity of the mind and specifically to uh the one that allows the this traffic going up and down um such as uh cognitive penetrability and there are still um a few people that um very are very engaged um in defending that kind of um of modularity of the mind that is um quite um agreeable and comp comptible with predictive coding and examples of that are for example more conservative um uh approaches within U predictive processing theories of the mind for example the predictive mind prediction error minimization uh mostly developed uh and headed by um by yakob Hui um that would be one that um you can find as as as having some agreement with with modularity of the mind of course the revised version of it which is uh the one in which uh it is it allows traffic to go up and down and that's of course you can see the parallel as it being predictions coming down prediction errors coming up and the traffic being very contentful in nature therefore you have a version that it has got this compatibility between modularity of the mind and um and um and the the predictive uh brain what happens is and I'll just wrap this up and go um then to active inference what happened there if you think about it is you let go of two things that were very precious for Jerry fer you let go of horizontal boundaries and vertical boundaries and once you let go of that you really have to ask the question what is left of the modularity of the Mind why should we keep it in place at all whatsoever right because what happened was you let go of one there was the basic pillar of of the modularity of the mind which was the boundary that was a distinguishing or dividing uh cognition on top and modules at the bottom right this boundary doesn't you don't have this boundary anymore because now you have predictions coming down so you let go of this boundary so Jerry doesn't get that and they have the vertical boundaries as well which were very precious because those were about information encapsulation such that different modules are the main specific and they do not talk to each other right so what you is you get rid of those vertical boundaries there's there's an interesting there's an interesting historical point there um that just sorry to interrupt but this is an interesting there there's an interesting chronology here which I actually learned during my undergrad where when I was looking at whether the language comprehension was fundamentally modular so it was related that's hence why I brought up n jsky in 83 fodal has very distinct definitions of information encapsulation and Dom specificity so do specificity is actually what we just spoke about that there's a proprietary database that each module works on right so visual the visual module work on uh photons or whatever you know or the language module works on syntactic nodes and information informational encapsulation in 83 was top down beliefs expectations doesn't affect the workings of those modules interestingly in in 2000s so he he writes another book or or article about modularity in the year 2000 he kind of conflicts and again I don't want to speak ill of Jerry fodo he's very important and he's also not here anymore so he can't defend himself but he seems to confuse information encapsulation with dom specificity because he talks about like this is a quote that a system is encaps a module is encapsulated if it's restrict if it function is restricted to what its proprietary database contains so I think what's interesting there is factly what you're were talking about in terms of horizontality and verticality kind of abandons the verticality there right he wants to uh yeah he he abandons the verticality in the sense that by by sort of a mission he concedes that maybe we should let go of this idea that top- down beliefs expectations don't influence lower levels he does keep the um the horizont horizontality which is let's keep these as distinct systems working on discrete um languages or data for ones of a better ter so so it's just an interesting historical fact there that I think if he was H frodel might um he could obviously offer some clarification but may also go back on his original 83 position of top down um the the restriction for top down information to flow into modules yes absolutely and um I think that this is also how you are absolutely right um in in in specifically the point about the conflation between information and caps olation and the main specificity these are two different things and one of the things that are that one of the aspects that I think would would need to be abandoned and that we abandoned in our Mark of blankets in the brain is precisely that notion of information right so because you can have domain specificity without relying on this notion of information encapsulation and things not conversing with each other because again to my perspective that is anthropomorphizing the brain and brain activ ity right we're talking about Dynamics and activities of the biology kind in the brain and there is an anthropomorphization uh that I don't think it's necessary and again it proved itself unnecessary right um so um but but we learn so much from from learning the history of where these ideas come from and they have a place of course U and I think it's important for us to precisely learn where they come from such that we start um improving and advancing our understanding of uh the brain and cognition in different ways so there is definitely place for for for Jerry Fodor um and I think that this is the place is to get us into other other other uh advancements in the field and what um what we have been able to to to to to see and uh through scientific evidence in in in neuroscience precisely that that kind of way of thinking that may have started as a metaphor doesn't really hold right and that there may be um domain specificity I think you put it in a very nice way there may be domain specificity without it being information encapsulation right and once you start having to let go of all of these basic pillars of modularity it's gone we can try to salvage it and then call it something and brand it something else but it's gone and that's when then you also see in the history well we have now new um computational uh capacity or strategy such as parallel distributed processing what shall we do let's try to apply the same thing that we did which is what if we would think about the mind as if it were a parallel distributed processing system and then you have the networks and predictive processing theor start rising up on the base of that analogy as well I am careful about going over and abond and above the anal the it being a metaphor right because I always think about the brain as a very biological kind of system as opposed to a full agent with full capacities that thinks and reasons and models as if it were a scientist so I push back on that I'm careful about that so the way from that position on then the way in which we theorize Mark of blank blanket is being useful was you start from those understandings that the brain is a dynamical complex system right that it changes over time that the ways in which a full agent engages with the environment is going to have significant impacts on the ways in which pathways are strengthened or are weakened in the ways in which the brain is going to develop itself as an organ right so you start from that very neurobiology kind of understanding of the brain that it is a complex system meaning that it is a coupled system it is situated in a body it is situated within other organs uh in in in the organism and this organism is situated in the environment right so we we we completely abandon the view of the brain as being a predictive machine there even though you can and there's lit chore on building upon the idea of the brain which of the brain being a predictive machine which comes from from Helm holds and more recently you can theorize it and I've been um I've been presenting on that as you you can theorize it as the low road or the high road to active inference which we can talk about later but the view is you can take that pathway that's not the one that we took uh we took the pathway of the very neurobiology uh kind of way of the brain being a highly situated organ within a larger system the larger system being situated in an environment and all of these influences and interactions are going to have an effect on the state that you're going to find the brain in a specific uh uh um point in time and space so so then um starting from that understanding as opposed to the brain being a a machine uh right so then we ask the question of whether or not uh because this is what this is crucial this is the crucial point which is why I started by by describing mark of blink is is this scientific construct that you can apply skill free comes skill free uh that you can apply to anything that is a thing there is an open system you can apply it you can also apply it to the brains but then you once you do that you have to test it out right you have to look into okay does this really make sense can we find patterns of activity that would be explainable through the application of Mark of blankets at different levels of organization within the brain itself so then in that particular paper what we did was we we are we start from a standpoint where we reject the modularity of the mind because we do not think that there are reasons scientific reasons to think that the the the the the brain is organized in very strict rigid ways call it modules whatever we want to call it but in very strict rigid ways we do not think that that's the case because there is scientific reason not to take that path and the Brain being a very Dynamic and changing um system then the question that you have is how do we make sense of it right how do we find patterns in such a dynamical system in such a coupled system how do we find those patterns can we find those patterns so then let's apply this um uh Mark of blanket formulas at different levels given the data that we have coming from DCM coming from dynamical causal modeling of the brain so basically um in very very simple terms The View here is that we put the brains in Scanners we get data out of the scanner and then we look at that and just vast data the comic sense of it Con and start to think about how to make sense of that you employ DCM as a model to make sense of that data and what you would get get from that is a nice causal topology about how um uh uh how connections are made and Dynamics are formed so you get a nice pattern of that if the model is is is is suitable and useful then you that's what you get so then from that kind of like um scientific um point we we we grab all of that all of those nice models uh from DCM and we see we look at whether or not we could apply the mark of blanket formalism so the four the two sets of States um internal and external active and sensory States in order to uh get an explanation of those pattern Dynamics at different scales so we start off with the the scale of a single neuron and then we have a cical column and then we have regions right and what is the what is the the the take on message from this because of course if if anybody's interested uh in the details then the paper is out there um but what is the take on message there is that we do find the same patterns of behavior across scales by virtue of employing this uh formalism and that is that is really good that's already quite um quite quite quite important because that means that you can get what we started off our conversation with you can get um understandings of activity that is occurring in the brain at different scales where uh each of those scales is both operationally closed and thermodynamically open so then what you get is you let completely go of that rigid systems that you had with the modularity of the mind you completely let go of that and what you get is an understanding of how um certain neurons in in in specific um environments you can think about it in a very simple way as like a cell in a Cell tissue how a cell in a Cell tissue is interacting and engaging with the tissue in order to do what it needs to do um so you can think about you can use this formalism to precisely understand how that particular system is behaving according to the environment that it is at so then you find these patterns of behaviors of like certain neurons firing given certain um conditions and then you get all this nice understanding of discou Behavior being situated that is highly Dynamic and it really is dependent on the environment that it is situated at what you also realize is that there are certain parts of the brain there are certain specific parts of the brain um that you can see that they may be insulated that they they they do not um conform to um active inference they do not exchange um and that's in a way if you wanted to say that would be what Jerry F had in mind as a modularity of the mind you could say that but all that happens there is a very simple bi biology process which is the system already knows what to do without needing to know what is going on in the environment it is what we say as it is set by Evolution it works so it's been working so well that way that it doesn't need to know or get to know feedback from its environment right so those would be the ones in which you would say oh it's encapsulated but it's informationally encapsulated I wouldn't say so right right right right yeah there's a lot of that there's a lot of that um cool I mean it I you also gave a really nice um description of for cognitive science there so exactly as I said in my introduction um the four e for those who can't remember embodied extended see if I remember enacted extended you get all of them yeah um and I'm enjoying the history lesson as well so um because I do as as you said I think the chronology is really important here just to get a picture of the philosophical foundations on which this work is being done but eight years after Jerry Fodor releases the modularity of the Mind the very important book comes out which is the embodied mind of Raa Thompson and rash before we go there because you work I guess within that framework of as you said for cognitive science the embodied mind I think listeners will be curious about you you mentioned scientific evidence refuting uh modularity I just wondering if you could provide this maybe just with one example of a module which is not exhibiting the typical um the typical features that Fodor laid out which would make it fundamentally modular so what what makes it open let's say to its environment yeah one very clear example comes from the all the research in neural reuse so that would itself already um sort of like move us away or it's it's I would say it's it's strong reasons to move away from that rigidity of modules that are do not change that are um as Jerry fod would conceive of it they are hardwired in that way so another feature that you lose is the automaticity right so the idea with the auto automaticity is that modules in a way are silly right they don't know really what's going on they are hardwired to automatically process the information that they domain specific to right so that's the automat automaticity they don't really have a say in the meta so to speak um even though I'm completely anop anthropomorph ing this story as as one would with the modularity of the mind uh so you you do lose that um with with with the the the large evidence on on on um on neural reuse where you see um which is extremely compelling and it it's absolutely fascinating that you see certain neurons that we used to be neurons for vision to become neurons for something else like like odit Tre system for example depending on the necessity of the whole overall system so what you see is the neural reuse um uh cases where well it really it really challenges the basic aspects or properties that would make a module a module right don't make domain specific or is it domain specific if it becomes the main specific to something else uh the automaticity um as well so the the the the neural reuse um I think would be a it's it's a great challenge um another one is of course neuroplasticity so the studies in neuroplasticity do um challenge again the rigidity of uh the the the the the non changeability of this hardwired um system yeah absolutely okay great that's really useful yeah yeah I always think about um I think I got to about the murk effect when I was learning about modularity of mind which is this auditory visual illusion that when people yeah people can search up or do the murk effect but again that that brings into question um this encapsulated idea that muls can't speak to each other okay excellent well as I said what became very popular uh after these kinds of rigid computational accounts of the Mind was an activist embodied models following verel

Thompson and rush and verel was of course a biologist and so um and he introduced the term aesis which kind of means this self organization that we now speak of or self- evidencing um before 1991 so so as you were saying we are inclined now to look at the brain as a biotic thing as a biological system and this is very much uh for grounded in um embodied cognition maybe it would be useful for everyone if we could just go through the four e firstly um it doesn't it doesn't have to be super it's a very rich history um so I'm you know as long as short as is suitable but maybe we could go yeah one by one and think about where is the evidence or where is he philosophy or as the intuition that this is true um and we can start I guess with the the first one which is embodiment what what does embodiment mean and kind of what is the research around the embodied brain what what does it focus on what does does it Center on all right very very good um so um embodiment is um something that is is a concept that comes from phenomenology um from the existentialist Philosophy from phenomenology it's a very rich uh concept and it is fundamental to understand a lot of confusion in the field because we need just we need to distinguish between weak and strong embodiment I will call strong embodiment the one that comes from chology and existentialist philosophy such as the one that for example Sim B had in mind the body that one has that grounds our experience of the world that's the kind of embodiment from SIM or um and um then within the E Frameworks uh within the E framework you have different theories and while they do agree with let's say a couple of um principles they are going to disagree age with within certain more nuanced aspects of their own theories and I think the embodiment is one that is mostly useful for us to understand the State of Affairs in the field and the theory and the confusions right so um let's say that it is very reasonable to to think that most people in the field of cognitive science today are not going to reject that the body plays a role to understand the brain that the body plays a role to understand cognition very few people would disagree with this right the followup question is how do you understand the body and I will get to that I will just say that only those that would conceive of cognition as encapsulated in the brain as a computer system such as the prediction error minimization accounts or for example yakob who is kind of conservative account um that there he says very clearly you can throw away the body the world and other people because the brain sufficiently as a machine takes care of the cognitive business so that would be the only account that I know of that would not bring in the body to play a fundamental role all other accounts do agree I think we have come to that agreement that the body plays a fundamental role embodiment plays a fundamental role now the question the follow-up question is how do conceive of embodiment and then you have two ways the weak and the strong weak embodiment is the one that is endorsed for example by um predictive processing theories such as Andy Clark as the Visionary uh behind it what happens there is that Andy Clark in his account invites the body to play a causal role in cognition and this is important because also with the with the with the with the with the body you can um interact with a world that is Technologic iCal where there are things that then give you the extended mind but I'll I'll get to that if if we have the time anyway so it's weak embodiment what does this mean this means that um cognition is a predictive processing that takes place in the brain so this agrees with more conservative predictive uh aor minimization accounts but not only in the brain predictions but the prediction processing machine includes the body what does this mean this means that the body is seen as part of the predictive machine the body is seen as computational cognition is computational so that's why this is weak embodiment so anic clar does not endorse the embodiment that MTI or Sim had in mind he endorses a very computationalist kind of embodiment so this is why you can kind of like to understand the E cognitive science you kind of have to spread it out within a range and then you get like um you get like the more conservative ones and the more radical ones right the more conservative ones are the ones that are going to say that cognition is computational the more radical ones are going to say that cognition is not computation and then you have a spread out of of um of the question that you can ask to help navigate the ecognition uh literature is how radical do you want to be because if you want to say that cognition is computational you're not radical at all right and radical doesn't have anything to do with original radical just means how radical you are in comparison to Old theories such as the the the computational theory of mind what is the computational theory of mind is everything we' been talking about with the modularity of the mind and predictive processing theories this is all computational theory of Mind how radical are you in relation to the computational theory of Mind where are the basic assumptions are to that cognition comes down to um uh computational processes of information processing that's computational theory of mind and that the mark of the mental is mental representation there would be more Nuance thing things to say but this would be two that you can ask yourself and then you can ask how radical am I away from these two assumptions right so that's the view and then you get like this weak embodiment which is very close to or if not it is um kind of like right in the border between computational theory of mind and ecognition almost like shouldn't be there if you would conceive that uh cognition within e the E umbrella is not computational right uh so that's the very weak embodiment and then you have the strong embodiment which is any theory that explains cognition by grounding cognition very much uh within the embodiment that the experience that one has of the world of course here what we talking about is uh the level of analysis and and understanding that uh people working in that field are looking at is um about the full um agent that is situated in an environment we are not talking about neurons of course we take that into consideration in the sense of the causal relations that I mentioned before which is how is it that the experience of the world is going to aect effect and influence the Bly biological processes and in there you also find the brain so that's kind of like the strong embodiment from mopon H and all of that kind of thing uh that we have and mostly applies to uh psych Psychotherapy and and and and psychological analysis and then you get into um uh let's say in activism right and then in an activism you find um different kinds again but let's say that both strong embodiment and in activism they reject these two things they reject that cognition is a computational process of information processing uh and they reject that the mark of the mental is mental representation so they reject these two things right they reject in kind of like different ways but non not non-compatible but they they they want to reject these two assumptions um so the an activist um has different different ways um in which uh you can also formulate of it um so you have the verela Thompson and rush so verela and Matana uh starting the autopoetic um uh in activism uh which is a very rich uh literature very based in in biology and extremely compelling uh and very well formed um and then of course um others more recently pick up on um autopoetic and activism and uh they have developed more recently um the framework towards uh sense making uh in activism now I want to just stop here to go to autois just to make the bridge with active inference where you can right where or at least where I find that it's reasonable to make that breach um autoptic and activism um brings out and a lot of recent work by an activists they also stress uh that out um they bring out the understanding of crucial aspects of living systems from bacteria to us um as having those two aspects as being thermodynamically open and as being operationally closed and those two aspects Lounge the system into a state of precariousness so these are main Concepts within theop poetic and activism and what this can then mean that we can think of is that what does it mean for a system to be precarious well what it means is that the system must must must interact with the environment in order to remain alive what does this mean well this means that the system is coupled with the environment now again what does this mean this means that the system the living being cannot be encapsulated from the environment what this would refute is the conservative computationalist views that cognition comes down to being encapsulated in the brain and we can throw away the body the world and other people so that's kind of like what this means so that's why the question is how radical are you from The View that cognition comes down to computations in the brain right how radical are you so we're making our way towards a more radical one and then once you find these ideas of like operationally closed thermodynamically open therefore a state of precariousness therefore must be coupled otherwise it dies then you say well this active inference framework actually captures that very nicely because I get conditional independency from internal and external stat that would allow me to get the operationally closure but I also get the direct influence of being thermodynamically open that comes from active and sensory States so then it really nicely captures that state of precariousness as a tool so that's that's nice I think and then you can um you can go as radical as you want which is to completely push back on um any views or kinds of theories or ways of thinking that would bring out uh cognition and cognitive systems from bacteria to more sophisticated inculturated systems like us um as um being computational of any kind and this is something that I've been presenting as augmented cognition in the in the sense of um the kinds of like computational models that we develop um and mathematical thinkings that we develop they are our cultural PR practices right so in that sense we developed these very sophisticated ways of making sense of the natural world of which brains are part of living systems are part of right we develop all of those very highly sophisticated systems and computational um capacity um to make sense of the world around us and what I'm very very careful is not to make that last step which would bring me back to the computational theory of mind to say that those computational strategies are ontological predictors of what the brain is of or or of what the living system is so I am very careful about that that maybe put me in the very radical position because I reject all of the computational theory of mind uh view um so that's how you kind of like make your way within the ecognition is by asking how radical are you in respect to to computational the of mine excellent great um I think we have two more e but we but we basically touched upon them um one is embedded um and the next one is extended um do I give a brief absolutely word to that absolutely so um the embedded one is the one that as I was spre spreading it out in range and I was saying that well um you have here very close uh you have a bound with the computational theory of mind and here you have the radical stuff so embedded is here close to the the the computational theory of mind and this is the E that I mentioned before when I explained that um very few people in com in cognitive science in the field would reject that the brain is embedded in the body and the body is embedded in the environment very few people would reject that so that's the embedded that's why this can be seen as almost close to the weak em embodiment that's the embeddedness very few people would talk about encapsulation only with that exception of the one that is the prediction rization by Yakov can throw away the world um and other PE the body and other people so that's the embededness is just the claim that the body is the brain is situated in the body the body is situated in the environment now then how you are going to explain that situatedness is where you are going to fall in different ease in the strong embod in the different kinds of uh in activism the radical in activism or the more sense making in activism um how you are going to explain that it's the embeddedness is what's going to make the difference on our radical you go and then you have the extended which I briefly mentioned um so the extended comes as a part of like a package uh with um uh radical predictive processing by and Clark so the package is that um it it is it is radical in uh relation and in comparison with prediction and minimization um or conservative one it's radical in relation to that but it's not radical in relation to other eats so what happens is that what you have is an understanding of cognition as being a computational process that takes place where well not only in the brain but also in the body right the body plays a causal role for that information processing predictive processing to occure and to happen uh in the ways that it does now it can also incorporate objects tools from the environment that are going to play a causal role of the same kind as um your body plays so then you have a whole system that is distributed it's is a computational system we still talking about computational theory of mind it is a computational system that involves brain body and tools and that's the extended mind great um and yeah I always feel the need to kind of refer people to where these ideas come from so um extend of my hypothesis Clark and Chara's classic paper um and definitely worth checking out I think when I hear uh these stronger an activist or stronger for recognition um perspectives which refute representation as the mark of the mental or Mark of cognition it makes me think uh where does Consciousness fit into all of this because as a conscious um you know subject uh being with a subjective start upon the world I you know I I have some sympathy for the enti list view because I think it feeds into some intuitions that I can't like this mind is privileged and can be divorced in some sense from the world upon which it does some computational work so if really we we we take the privilege away from representation um which which I think is also a kind of worm so we don't have to get into exactly what is a representation but maybe we maybe we can go there um what's what's the kind of R of qualia subjectivity intentionality and the kind of uh features of Consciousness oh thank you for that question I always feel like when um when we go that path um the mental representation being the mark of the mental um it feels like uh we are getting to watch a TV show halfway through like in season two right because we don't ask the questions where the representations and content comes from right so that's the hard problem of content so I know that we want to get that because that would allow us because we already are in the place where we do have me we can engage and develop and more and more every more sophisticated mental representations that what we are that's what part of what we are doing here is engaging in ways in which we are involving and employing uh mental representation such that we can understand each other right so it's like we're already half season halfway through this through the TV show we are already like in season three um so but there was a beginning of the TV show right well you didn't have any of this you as a baby as an infant could not be communicated with me in this way about these things where does that come from that comes from something called inculturation right so I think before before we start explaining things as if they were always like that we need to understand that there is a trajectory in cognition in cognitive development that we shall not forget so all of those things had to be acquired unless you want to defend nativism that that would be also could could be very close to Jerry F as well because you already have these starter packs that you already have this very very much in place um Machinery that would allow us you to have all of these sensory all of this semantic content already ready to use that you could start making sense now if you reject folder you reject it all now you have to answer the question how can we do this where does this come from you've rejected folder right so you have a story to tell now the way that I I see it and many um in activists do see it uh and by the way also very much compatible than with mathematical uh Frameworks of dynamical and complex Theory um and that is that is employed in in in a lot of psychology uh uh Empirical research as opposed to the computational theory of mind uh even though that's the mainstreaming in cognitive psychology it's not the only one um and the way in which you can think about it is that you um cognition the m so what I'll start by saying this I've I've heard many times people ask me but how is it possible that there's cognition but there is no no mental representations no one is saying that so that's not a claim that's not a claim from an activist an activist an activists don't say that there aren mental representations what they tell what they want to tell you is that mental representations do not come for free you are not born with them you had to do things in the environment in order to acquire them you have to to develop it as a skill so we give you a story about a hard problem of content where does that come from when it does not come as within the nice starter pack that cherry foder or decart would tell you that it comes from so once you get rid of that you have a story to tell and the story is that it depends on inculturation so basically as you start as an infant as a baby we we come into the world and we start interacting with the world this world is and this is the key point this world is already social it's already a social cultural setting there is no baby that is born and is developing within a setting that is not social already so it is in the ways in which the baby interacts with the their caregivers that the parents are the caregivers and the world around them that is going to gradually allow them to build um simpler um Concepts simple ways of using Concepts and then scuff holding that into a a numerical system in a conceptual framework and then later on in being able to engage in logical thinking and reasoning and then being able to express some sophisticated ideas and maybe even becoming a scientist and contribute to developing highly sophisticated computational models that study black holes so it's an inculturated process of course as we are here doing this interaction um what we are using is we are making the best use of our most skilled to so far uh and I'm sure that um that will go keep going in a in a in a a trajectory but our best skills of um explaining ideas and discussing so we are using and making the best use of our best mental representations but I also have to say that this interaction does not reduce to those U mental representations there is an embodiment aspect of the situation that we are at a a very clear uh example of that is that the interaction is different whether or not we are having the interaction in person or we are having the interaction online we'll make things different because we don't share the same physical space so this is just an example of um how embodiment here would play uh uh uh some some would play out but the idea is that eventually you'll start developing the capacity to develop and engage in mental representations also as you talk to other people and you make sense of the world around you okay cool um I'm going to play not going to play Devil's Advocate but um and I also don't want to make you commit to any position nor am I asking you to solve the hard problem of Consciousness because that's not that's not going to happen this also not the purpose of the podcast but I understand in that sense that mental representations are kind of Downstream on enacted behavior on social cultural norms right it they as you say there's a lot of evidence to suggest that it's not some hardwired Chomsky and module right there we can take that as kind of axiomatic I guess my question here is that kind of makes sense in terms of mental representations per se let's say they develop out of this inactivist um way of being that we have what it doesn't necessarily answer is the kind of characteristic features and the idiosyncratic features of Consciousness so the actual like fundamental what it is like to be me what it is like to be you to use Thomas Nagel's terms so I guess my question here is um again not trying to pin you down I understand there's a lot of nuance here would the an activist be uh sort of would they be attracted to the notion that

Consciousness is epip phenomenal and by that I mean for people who don't know epip phenomenal Consciousness is the idea that Consciousness is like the steam coming off a steam train so it's caused by uh neural physiological Behavior but doesn't actually have any causal effect on those um on those mechanisms so it happens it's there right and it kind of depends on what philosophy of Mind Theory you um subscribe to but would they say that Consciousness has no causal role um I'm imagining that there aren't many substance dualist in activists but what what what kind of considerations have they made of the role of Consciousness right um so I I I'll just bring some some something in so that so when we talk about qualia uh epiphenomenalism um um and uh what it is like um like in the bat um what is like to be a bad Thomas nle when we are or or the the hard problem of Consciousness when we talk about that we are within a certain conceptual toolkit right we are already within what wienstein would call a bottle a theory right and it's important to to to understand that when we start making our way within a theoretical path it's like we're building blocks of a house and then we getting a certain kind of house by virtue of the the belief system thoughts that and the path that we are doing that's the kind of like conceptual Machinery that you get once you play that game so and by playing that game I'm here referring to or having in mind viin Stein's language games so you can tell the story of or you can do the epistemology of understanding uh how cognition comes around and about and works by playing different games and that is the traditional analytic philosophy of Mind Game and that game is very associated with the computational theory of mind so what I'm trying to say here is that once you put on certain goggles there are certain things that you're going to see and there are certain things that you cannot see so once you start on that path of thinking Journey those are the problems that you're going to find those are the problems that like the hard problem of Consciousness is a problem that you're going to find because then what you get is uh how do we study both the easy problem and the hard problem so you necessarily when you start going in that direction you find those problems and then you have the explanatory Gap that how we close the gap so these are problems that come up within the computational theory thinking language game so those are the goggles right and of course we can try to make these theories talk to each other but it's very important to understand that it's almost like axiomatic in the sense that you start solving an equation in one way it the equation will look a certain way you start solving the equation in a different way the equation will look in a in a different way right so that's almost what we are talking about right because we are not starting from the point in E cognition doesn't start from the point of view of the brain in the other um more traditional computational theor of M of course it does so once you do start from the bra brain being machine note what questions come up okay so if the brain is a machine then how is it that it gives rise so it's a scaling up problem how is it that it gives a rise to Consciousness how does this machine give rise to Consciousness because there are there is evidence that there's Consciousness that is what it is like to be about there's qualia there's a hard problem of Consciousness this is within that kind of thinking because you start by the brain being a machine and then you ask how do we get to the good stuff and then you get all of that from within the more radical e cognitive science you don't start from the brain being a machine that's the first thing you reject right and that's also in dynamical and complex systems theory the case you have phenomenal work by Van Helder and other people that precisely reject that cognition has anything to do of being anywhere or we should use any conceptual uh any concept from a computational scientist perspective to Kit right so then you don't start from there you start from the point of view that living systems are precarious and they engage with the world to maintain themselves the brain is part of the story but you don't start there right so in this engagement what happens is you have you engage with the world because you have certain goals usually the goals have to do in a more minimalistic biology way is that you don't want to die it's the precariousness of of pushing back on the second law of Thermodynamics right so you engage in the world for that reason because you want to remain alive typically this is the most basic biology there's much more than that but let's just stay with that so then what happens is you start exploring the world and in exploring the world you do that by exploring your body so that's when the body comes in as the body is the means by which you have access and explore the world and respond to the world's and invitations to exploration so it is in these interactions with the world that are extremely embodied and limited they're both constrained and possible by the body that whatever you do it's cognitive behavior and this cognitive behavior is what are the goods that we want to understand in cognitive science and we do that by by explaining them through patterns of behavior right so it's completely different goggles yeah I I am going to push back because I think it's I think it's good it's not adversarial it's collaborative um I I just sense or I just kind of believe that the Vicken styan relativism has its has its limits right it has its points where you say well there are certain things that maybe aren't relative in the language game and maybe one of those are the fundamental aspects of my experience that I can attest to regardless of if I'm in a braid of lat or if I'm an evil demon is tricking me right like unless I adopt some illusionism about Consciousness which I'm not willing to do but again like it's not there there's not a viable position to do but I'm going to take it for granted that like even even radical illusionists wouldn't say that qualia doesn't exist it's just not what we think it is and I'm think here Kei Frankish and den it so for me like the Vick steinan relativism that qualia is itself part of a um Paradigm to use a sort of term of Thomas or it's part of a boss or it's part theoric yes in the sense of the linguistic aspect of it but the fundamental phenomenon is at least for what I can tell is real so what is the alternative I guess is my question right yet I I I don't I'm not refuting the notion that a cognition is in some way constrained and opened up impossibility by having a body by being embedded in the social context and so on but we still have to GR in my opinion we still have to grasp with the fundamental facts of phenomenal experience so I'm sorry if I sound like David Charmers and David Charmers parrot but no it seems to me that like I can't get away from the explanatory Gap and I don't think that that's a problem in my theoretical framing of the problem I think that's because the phenomena phenomena associated with Consciousness is so radically different from anything else in the physical Universe yeah and and there are very um I must say that there are very good points that are made by David chalas in that respect that I that I uh very much um agree with um now to go back to that and to link that in a way that would be um that would be compatible within uh my view of an activism I would say that and also wienstein um there is a real world that you don't have access to so I completely agree with what you said right I completely agree uh that there is a limit to relativism yes that's the limit the limit is the limits of my view I'm not going to say the limits of my language as which I did in the first in the first book in trus because you overcame that so I'm just going to um I'm going to dare to reformulate that for him and say the limits of my perspective are my world so there's there there there are aspects of the world that I will not have access to and that's just the way things are I will not have access to your experience of the world because and here's the inactivist bit coming in because my experience is determined by the set of past interactions that I've had with the world and this moment right here where um this experience is taking place and everything else outside of this I don't have access to that's the bit in the Bo in in the box right so the the the idea is not the rejection of other perspectives and that's precisely vien Stein's point is understanding understanding that there are it's the humility of understanding that there are other perspectives in the world and you do not have access to them you only have access to your own uh very um specific kind of perspective to the world now another thing that he adds that it may be uh useful um is to say that for him these are seen in terms of aspects of the world you see the world through the aspects that are made relevant to you right here right now given what given your past experiences your past interactions of the world that's why certain things will become Salient in this particular moment in the environment which connecting to the other Theory you would call the what it is like to be me to what it is like see I don't know if I would um the well to me I wouldn't right I mean the the salience of my environment I completely agree is shaped by my body it's shaped by my homeostatic requirements it's shaped by my sociocultural upbringing status and so on but the fact of the matter is is that things we could describe things as being Salient without necessarily invoking well that's a good question I guess whether we could invoke salience of out Consciousness but I guess I I mean as a force experiment you can in a sense um I had John Vu on last week and he was talking about relevance realization and so it's very sort of still fresh in my mind I said my my reading of this would be that salience is just a manifestation of the conscious experience itself but are we going to get any closer to the actual fundamental characteristics right like whether it's salience or visual perception or auditory perception I mean like the Charmers problem is still there right uh the Chara's problem is okay let's let me just go back to the initial thing because otherwise I'll lose it um that you said um um the the note that the difference between the Frameworks is quite it's quite this simple thing which is Things become relevant and Salient and experienced in a certain way within ecognition because of the embodied experiences that you've had in the past within a computational theory of mind you might say the same you might say things are relevant what it is like to be me why well because of the way that information has been process processed in the brain as a as a as a machine so these are different explanations for the same kind of phenomenon they're diff they are completely different so this is where they are aart right um now as to the problem of um the heart I think when you said Chara problem you meant the heart problem of Consciousness right um y okay so it may be the case and this is also a Vian sign move so I'm not going to take credit for that um uh it may be the case and this is very in line with with David chers which is why I think there are certain features of it of of of chmer thinking that I that I really like and agree with um vians makes this point not everything that is there to be understood about the mind can be understood through the scientific meod and this is the hard problem of consciousness that chmer talks about right and and I think that he makes a good point there I think that he is very um cever and very very correct and I completely agree with three points that he makes which is that many people are falling up on the Trap of claiming to study Consciousness and experience and a first person however you want to call it depending on the theory you're coming from um embodiment if you like they claim to study embodiment and conscious experience and all of those kind of like good things that really make us by going to the brain Alone by studying easy stuff and that is a theoretical trap it's a fallacy and that's where I really agree with with David charma's work which is why I'm doing what I'm doing in in ecognition precisely because I do not want to get into that trap which I think it's it's a trap as well I agree with him great wonderful well that's our that's our convergence point and again I it's it's it's not so much that we need to solve the hard problem but I'm just curious about whether it retains its problematic nature in in for so that that that's good to get some clarity on that I Rend I wonder just it it's it's a difficult question but and again I'm not trying to get you to solve the whole problem I'm just curious about your thoughts on this I said I had John Vu on last week and he's got this idea that getting more precise on the f function of Consciousness would help us in resolving the nature of Consciousness the hard problem do you see any viability in that link you can get some something by the function if you within the Frameworks that I work on active inference and activism right Within These Frameworks I'm going to give the answer within these Frameworks you can get some explanatory traction you can get some um epistemic virtue out of thinking function but it doesn't get you all the way through it gets you to understand the precariousness and why systems behave in the way that they do it's because they the function is that they want to push back on the second law of Thermodynamics they want to remain alive but then you think about a point that Simon DEA made which is also a very clever point which is if all that is to life is to not to perish then what is it that makes us different to vegetation right so function gives you as much as it's interesting and relevant to the point that it explains or it gives you as far as a cross life explanation so one thing that you get is the function is all life does not want to die typically That's the Law that's the that's the free energy principle that's all but it gets you as far as that and you want more because you want to get to the goods of like uh embodied experience right why people behaving the way that they do right so here I think that you can bring in also again some on the above wise is is is I think is a good segue into um making this point which is very under the E umbrella but I think she makes this point in in a way that here is it's going to make the point um clearer which is that if if again going back to the point if all that is to life is not to perish then what makes us different from vegetation what makes us different from vation is that we are in um a kind of cognitive development trajectory that we become enculturated we are situated in Social cultural environments and once that happens we start behaving in ways that not only do we get the conceptual toolkits the reasoning Etc but we also have access and contribute and Co construct those social environments from the very embodied situation that we are at so in a way the biological body constraints the access and co-construction that you do with the world as well as the social culturally developed embodiment also plays a role in the participation that you have or not have within the environment that you navigate now she puts this in terms of uh the living being between it's it's a condition between freedom and facticity which see how how compatible this is with what we were saying about push pushing back on the on the second law of Thermodynamics you want to remain alive but you don't want to remain alive within an encapsulated environment you want to remain alive and do really well in a social cultural environment which is why we are more than simple vegetation that just wants to leave we want to live well we want to live good we want to do things that are good for us so we start now entering the realm of reasons like reasons that we need to give for the things that we do in the ways that we act right so now we start to become very sophisticated start to get to this like uh mental representation if you will but it doesn't come for free it's not a commodity we we now we start getting into like season 3 right which is like things start to get like really interesting so then you can explain um embodied experience as being social culturally situated that is dependent on the embodiment that you have and that embodiment that you have is what is going to somehow determine where you are navigating continuously between freedoms and facticity and she puts it even more precisely between um agency and objectification and then you start reasons to how you navigate the social cultural World wonderful yeah yeah it's the first time I've had s b brought up in intive inference conversation but um she that that's great uh yeah it's a wonderful segue actually into existentialism and phenomenology and and we're big fans of mlo Ponte and haiger and draus on this on this podcast as well I thought what you said was interesting which is that as you as you said correctly the free energy principle is a principle it's like hamiltonian principle of least action right it's kind of a where it's it's a first principles ontological explanation of just like what in the case of the FY principle what things have to do if they're going to continue through time to persist but active inference um or in certain cases predicted processing or in certain cases computational theories of mind or in certain cases it activat theories mind are what we call Process theories so it's if we take the free agent principle just to be axiomatically true in the sense that it's kind of and I think Carl would say this himself it if you actually put it down to its nuts and bolt it doesn't say too much right it's things do what things have to do to continue to be the things that they are really and there's a lot of fun maths but it's really about how it's implemented in us or artificial intelligence or whatever it is um those process theories become falsifiable so predicted processing might not be the right account of uh continuous St space um active inference or um perhaps the mind is uh you know fundamentally Downstream on the body in in a radically embodied way I guess because it's falsifiable um a question that came to my mind when you were speaking um about activism is and the history as well of cognitive science is as you say there's very few people who will adopt a radical computational um position these days daycard John C for example but for cognitive science is very popular right to put it simply can you envisage a future where for where future science and future philosophy tell us that viewing the mind as embodied enacted embedded what was the last one go on Wikipedia um yeah is not the the right way to view it as we now look back on the computational picture and think it's incomplete yeah okay so I I'm going to go back to the goggles um it it seems that the ways in so so science is built on um on First Steps right on the goggles that you put and then you end up with certain kinds of um theories that now you have and now what would I do with this I test it I see if it makes sense right in with the case that it doesn't make sense uh we we we don't know but it is about putting these goggles um and a lot of what is done still in in cognitive psychology is very daycard is very starting in in season 3 in like okay minds are mental representation so nobody cares about things that or about experiments or setting up experiments or thinking or reasoning about how did it come to be right where we we should put more effort on that how it came to be which is what by the way dynamical in complex systems theory applied to psychology does that's all they do all they do is to uh focus on the developmental trajectories to understand how they evolve and develop how States change into um developing and evolving um and and much of what is done in cognitive psychology is starting in season 3 is the the mind is representational and now we got to find that not only in our cognitive psychology experiments as we put as we as we design our experiments right is with that assumption in place and that's why this is so dangerous it may be the case that one is right to think that the mark of the mental is mental representation but not criticizing it and not having a critical point of view about it what what it means is that everything that we do within cognitive science science within setting up experiments within setting up computational models simulations it's going to be starting with the assumption that the mark of the mental is mental representation and that the brain is an information processing machine so this is going to be defining for everything that we do and all that I'm saying is that let's take a step back and go to the beginning of the series as opposed to starting in in in season three and question these things and if they still make sense great we've done a very good job right so so that's that's the point of view right and I think that what ecognition the more radical view does is it takes away that those toys and puts you into okay when you have to start from the very beginning what do you have and how do you build so it it it puts things into into pressure on whether that is the case because when you do the bit of the history of philosophy of mind and you realize that a lot of it has been done by borrowing conceptual toolkits from computational science then you ask the question what is left for philosophy of mind is there a philosophy of mind or is there a borrowing conceptual toolkit from computer science now that is a that's a provocative question I'm not going to I'm not going to try and answer it it it it's cool the way you talk about the the scientific theories it reminds me we had ma alberon as our second guest and actually on the day of us calling this the podcast just came out so everyone should definitely go and watch it but we had a long chat about epistemic communities and the kind of silos that people end up in and that's actually very much akin to what you were saying that people can really end up stuck in their scientific Paradigm and I'm really hoping that doesn't happen to me I think it's kind of happening already I mean it makes me think the my my sort of um philosophy science is not as good as it should be and it should get better but I'm aware of Thomas C in this idea of the Paradigm the way it sounds to me is that like I don't know I'm adopting a kind of more heelan perspective where um you got the computational theory of the mind and in activism like radical in activism as your thesis um and your sort of antithesis and you have this dialectical relationship right and then what you might end up with is a symphysis and that might I don't know it's a very broad question but do you see science unfolding in that or is it kind of more of a incremental Pursuit towards a higher or or getting closer to truth so um yeah you do have those two different paradigms and I would answer that in in in in in a in a very clear way I think that all that we have in the computational theory of mind is our human social culturally situated experience as a human species best efforts to understand a phenomenon in the natural world so we develop all this computational Machinery strategies models Etc because we this is what makes us human we want to understand the world around us and part of the world around us is human beings right so then we we have all of that as a reflection or as telling or as a manifestation of what we can do as a species when we come down to being epistemic communities and these epistemic communities are and ma was absolutely right are completely social situated so uh science itself like our own individual experience of the world does not occure in an encapsulated manner we could aim for an ideal bird ey view perspective of being doing science in a way that is completely objective but as wienstein already told us we don't have that we cannot get out of the trajectory that brought us here we cannot have access to that other perspective only our own and that is going to be biased and overlooking that it's dangerous because one would claim that that that is possible and that's not that's why our best our best um capacity as a human species is these epistemic communities where we can exchange ideas test ideas with each other give feedback to each other and improve each other's thinkings and um an experiment such that we improve in a collective effort that is scientific wonderful I like that it's spiriting I don't like the idea that we're just stepping on the the toes of our ancestors maybe they did have something good to tell us and we can build together but maybe never get there because we're blinded in some sense I mean um obviously where one cannot speak therefore one must be silent um in this case we could go on forever um I do want to make it clear that your work is incredibly Broad and diverse and touches on a wonderful range of things so we could have spoken about religion you have a wonderful paper on religion from an act of inference perspective um autism among other things uh we didn't even touch upon artificial intelligence agents um so there's going to have to be around two I suppose it there's where can people find your work what are you working on what have you got coming out where can people find you because I'm sure they'll want to know more about what you're doing yeah just on my website that's usually what is updated um which is www. energy poo.com so usually there you got the links to other places like Publications Etc and on Twitter as well I'm usually active so anything that comes out uh any uh conference that I'm organizing participating or papers that are out I always share them on on on my Twitter so that's usually the places super great okay that's wonderful well it's half past 11 UK p.m. so I need to need to dash off to bed but this has been an absolute pleasure um it thank you so much for your time and yeah as I said we should definitely do this again oh thank you so much what a terrific uh exciting uh uh conversation thank you so much and definitely up for round two well thank you thank you